Aircraft Engine Remaining Useful Life Prediction

NASA C-MAPSS FD001

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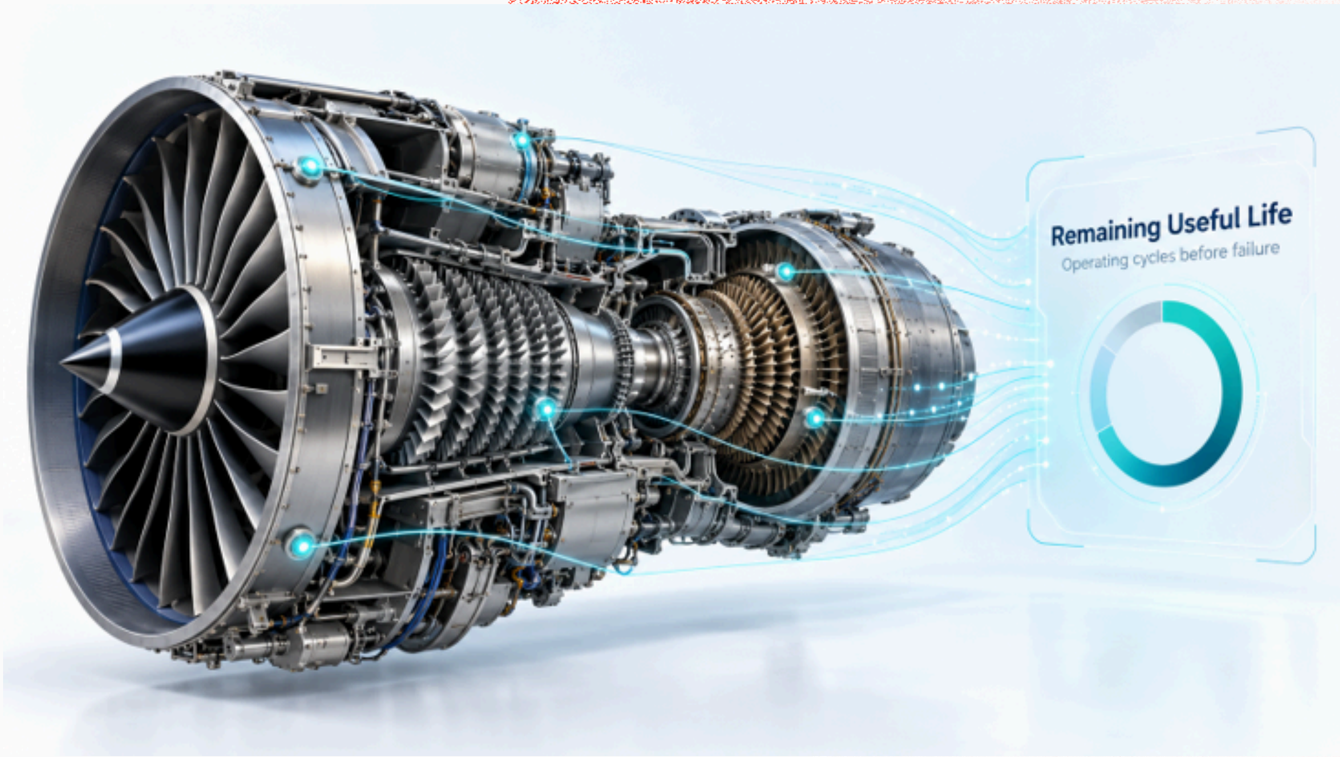


Table of Contents

The problem	1	The Models	5
The Workflow	2	Streamlit App	6
The Dataset	3	Conclusion	7
EDA	4	Roadmap	8

The problem

Goal: estimate remaining operating cycles before engine failure.

Training data: 20,631 rows from 100 engine trajectories.

Inputs: cycle, 3 operating settings and 21 sensor channels.

RUL = final training cycle − current cycle.

Example: Engine 1 at cycle 1 has RUL = 192 − 1 = 191.



2023

CYCLE



3

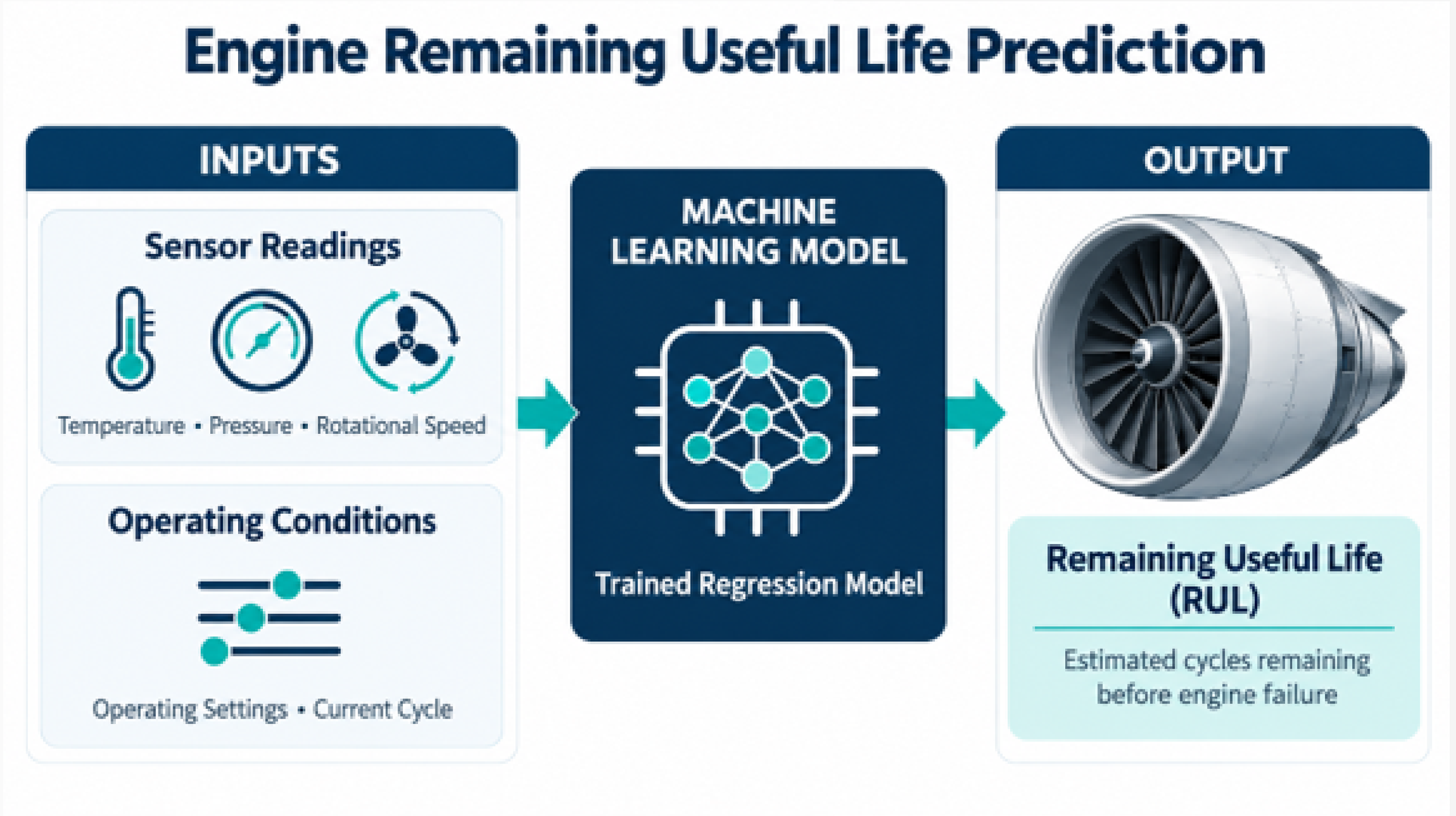
PERATING SETTINGS



21

SENSOR CHANNELS

The Workflow



The Dataset

NASA C-MAPSS DATASET

Simulated aircraft engine data containing engine IDs, operating cycles, 3 operating settings, and 21 sensor readings. Each row represents one engine at one cycle, used to predict Remaining Useful Life (RUL).

	engine_id	cycle	setting_1	setting_2	setting_3	sensor_1	sensor_2	sensor_3	sensor_4
0	1	1	-0.0007	-0.0004	100.0	518.67	641.82	1589.70	1400.60
1	1	2	0.0019	-0.0003	100.0	518.67	642.15	1591.82	1403.14
2	1	3	-0.0043	0.0003	100.0	518.67	642.35	1587.99	1404.20
3	1	4	0.0007	0.0000	100.0	518.67	642.35	1582.79	1401.87
4	1	5	-0.0019	-0.0002	100.0	518.67	642.37	1582.85	1406.22

5 rows × 26 columns

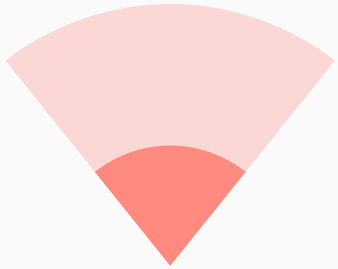
The EDA



INITIAL
EXPLORING DATA



EDA VISUALIZING

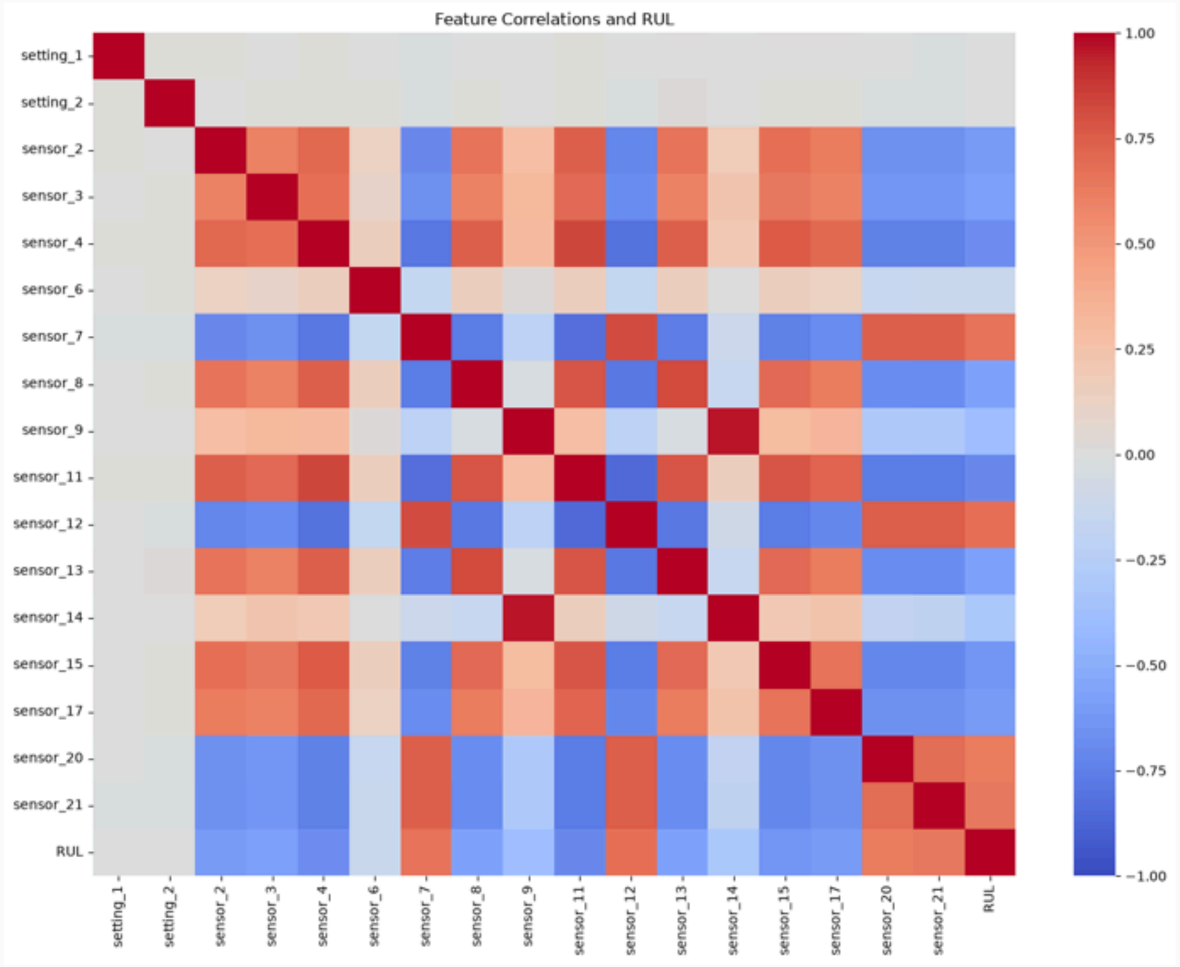


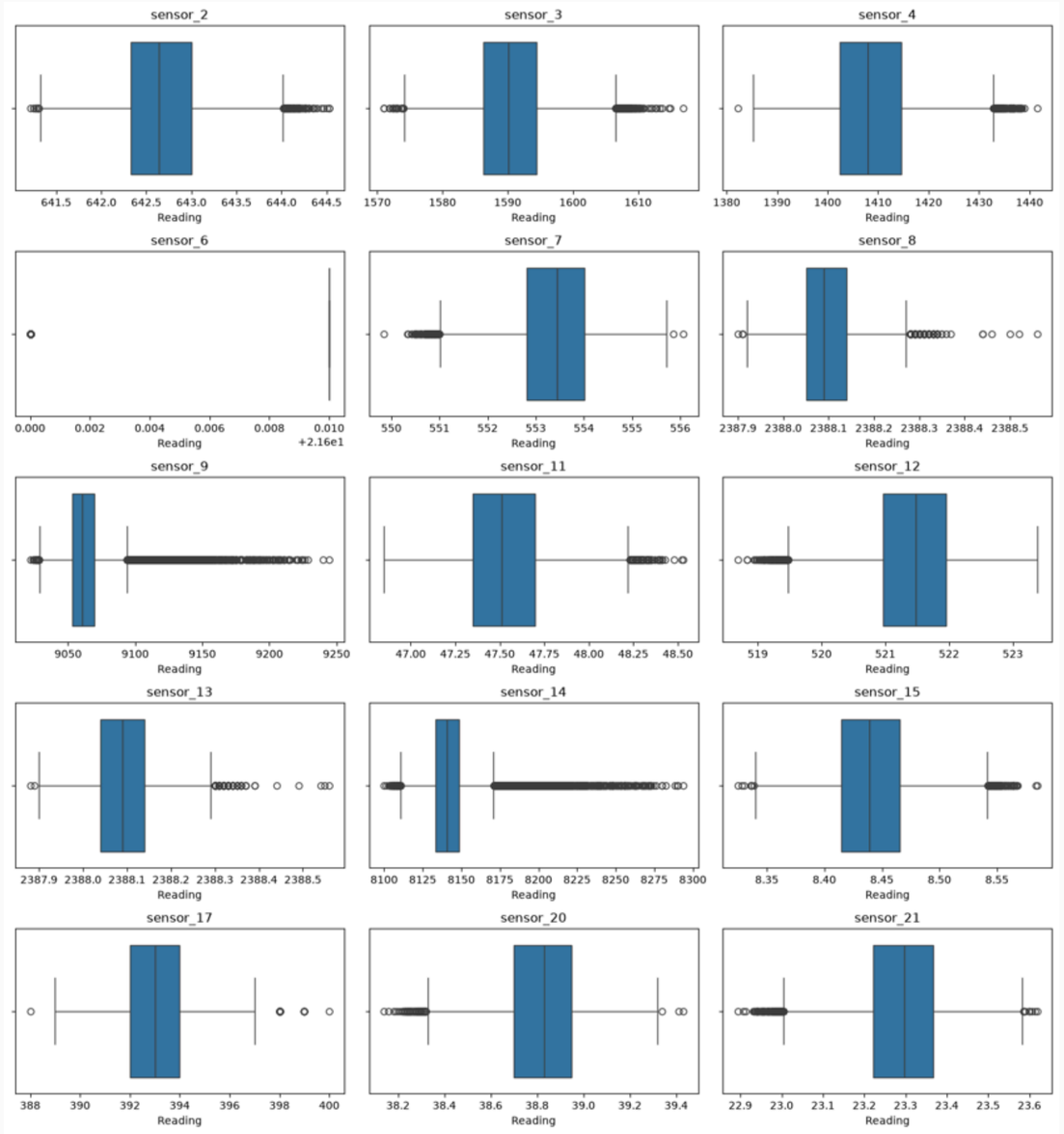
DECISIONS

Initial Exploring Data

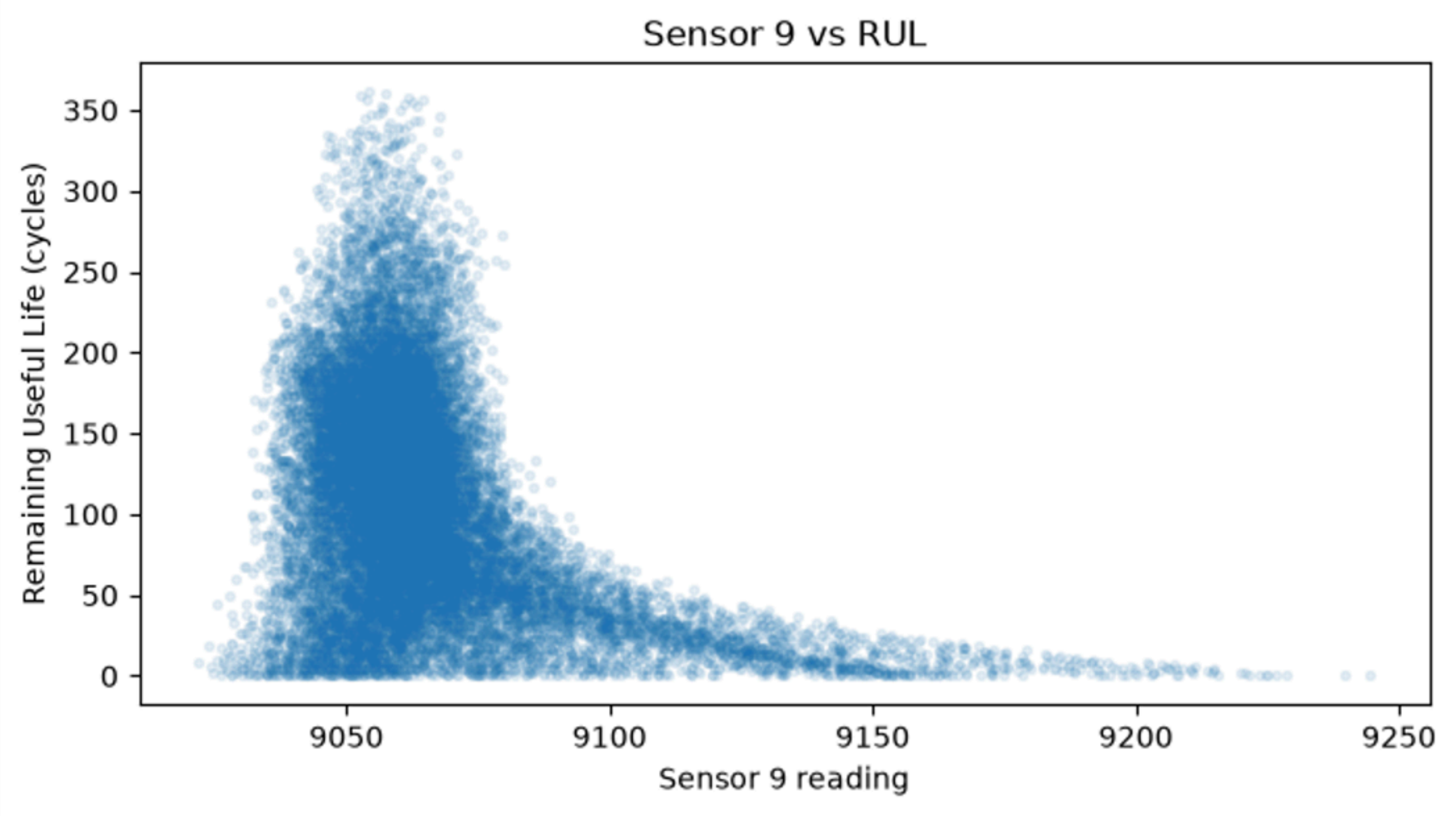
Item	Summary
Total records	20,631
Total columns	26
Missing values	None
Data types	22 float columns and 4 integer columns
Column names	Initially numbered from 0 to 25

EDA visualizing

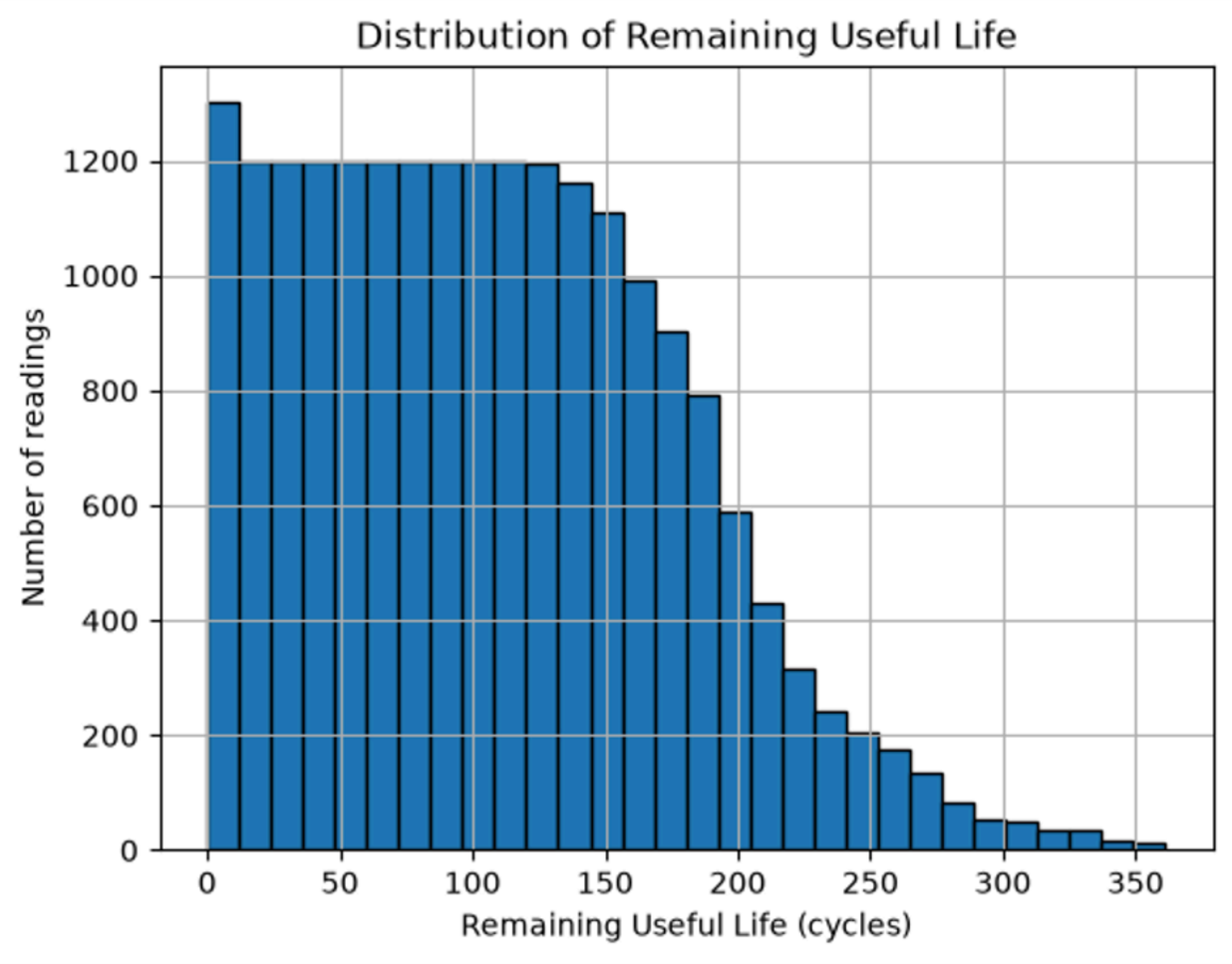


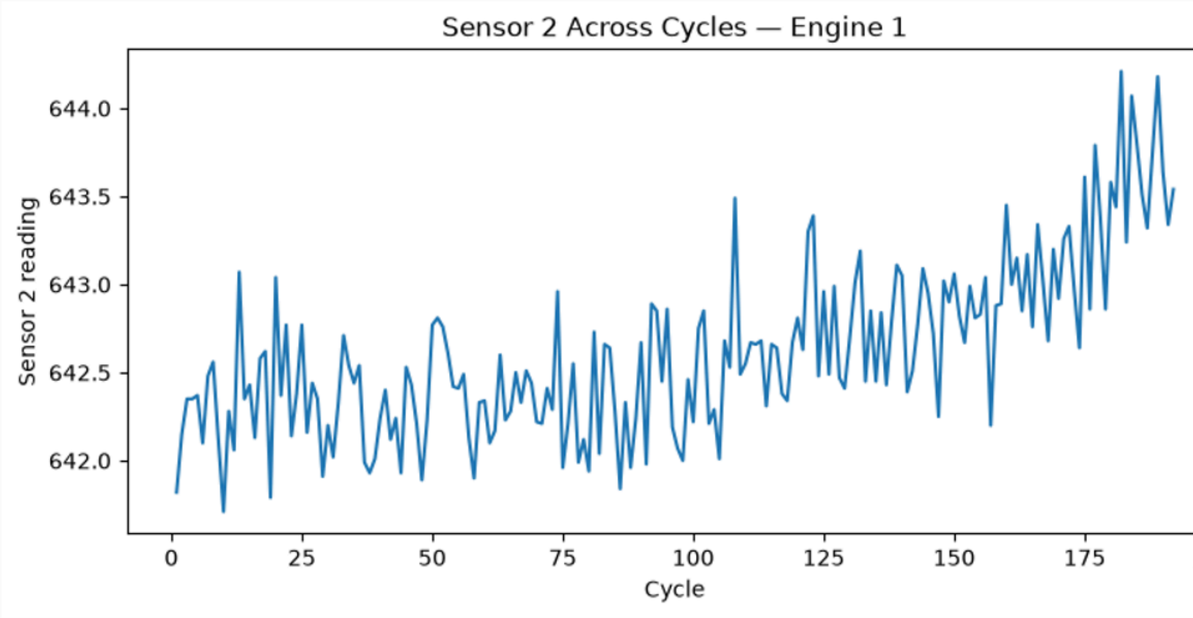


EDA visualizing



EDA visualizing





Finding	Decision
No missing values or duplicate rows	No imputation or duplicate removal
7 constant features	Exclude constant columns
Statistical outliers	Retain without evidence of error
Numeric measurements	No categorical encoding
Different feature scales	StandardScaler in linear-model pipelines
Engine identifier	Use for grouping, exclude from predictors

The Models

Model	MAE (cycles)	RMSE (cycles)	R^2
Gradient Boosting	18.831	25.226	0.632
Random Forest	19.139	25.606	0.62
Ridge	25.975	31.25	0.434
Linear Regression	25.975	31.25	0.434

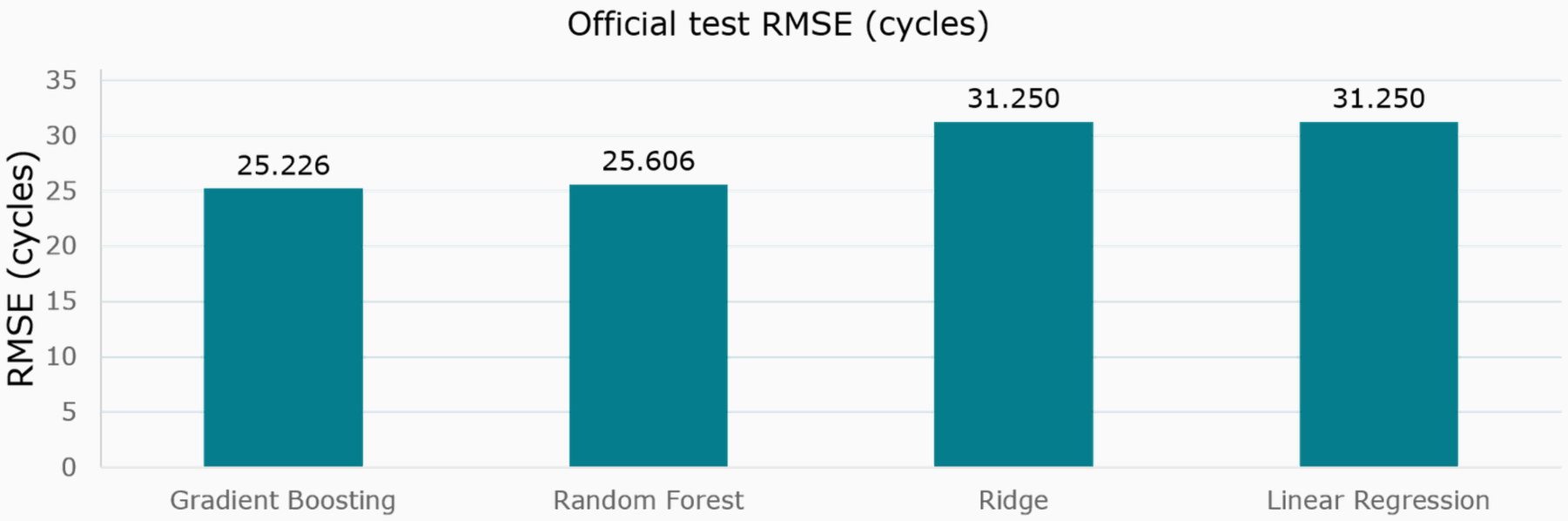
MAE: 18.831 CYCLES AVERAGE ABSOLUTE ERROR

The Models

Model	R^2 — Full RUL	R^2 — RUL capped at 125 cycles
Linear Regression	0.434	0.717
Ridge	0.434	0.717
Random Forest	0.586	0.823
Gradient Boosting	**0.639**	**0.837**

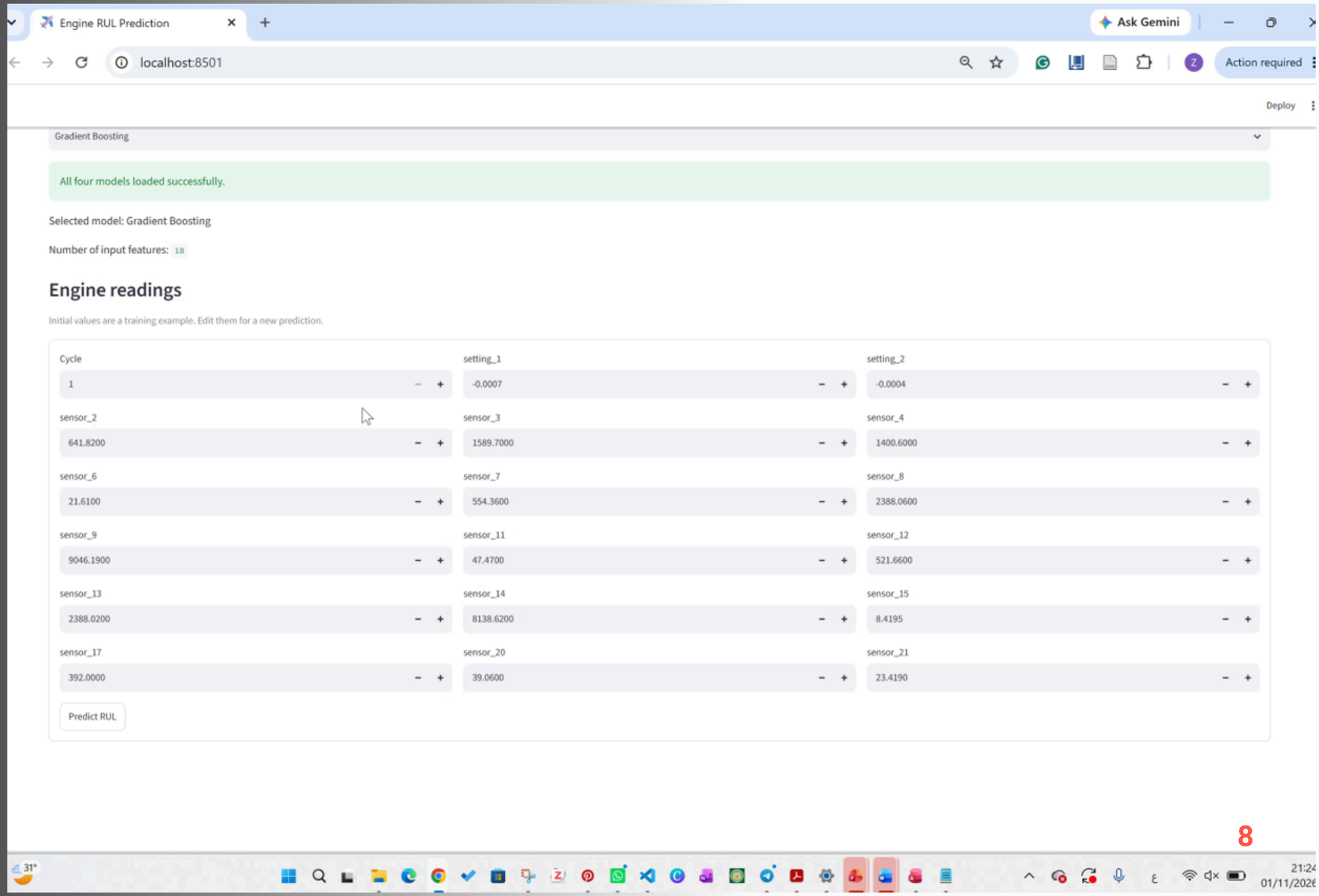
BOTH EXPERIMENTS USED THE SAME TEST ENGINES. THE FULL-RUL EXPERIMENT USED THE ORIGINAL TARGETS FOR TRAINING AND EVALUATION. THE CAPPED-RUL EXPERIMENT APPLIED A 125-CYCLE CAP TO BOTH.

The models



Gradient Boosting reduces test RMSE by 19.3% versus Linear Regression

STREAMLIT APP



Conclusions

- Built regression models to predict aircraft engine Remaining Useful Life (RUL) from sensor readings and operating settings.
- The project illustrates how RUL predictions could support maintenance planning.

SUBSCRIPTION-BASED LICENSING

Conclusion: Gradient Boosting achieved MAE 18.831 and RMSE 25.226.

LIMITATIONS

Predictions may not generalize to different engines or operating conditions.

FUTURE IMPROVEMENTS

Evaluate the model under broader operating conditions and on data that was not used for model selection or tuning.